

Mind the Gap:

Understanding, Defining and Handling Missing Accelerometer Data



Mia Tackney | Health Research from Home | Hackathon

Hello!

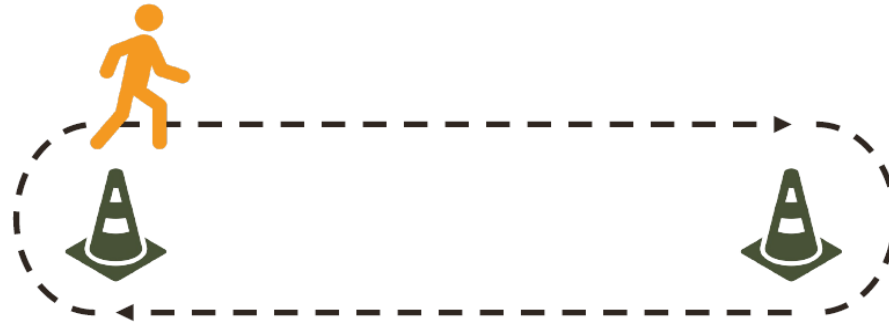
I am statistician who works on methodology around digital outcome measures for **clinical trials**. This talk will focus on:

- Digital outcome measures in clinical trials
- Missing data in Accelerometers
- Handling missing data on the epoch-level
- PPIE perspectives on missing data

6 Minute Walk Test

This is an outcome measure used in clinical trials in many different disease areas.

6 min



- Requires hospital visit (can be a high burden).
- Snapshot at one point in time – can be affected by how tired or how motivated the person is that day

Examples of Digital Outcome Measures



Digital walk test

- Participants can choose location
- More frequent testing possible

[Robertson et al (2024). Am J Respir Crit Care Med]



Accelerometer worn on hip/wrist

- Passively measures people's physical activity throughout the day

[Harris et al (2017). PLOS medicine]

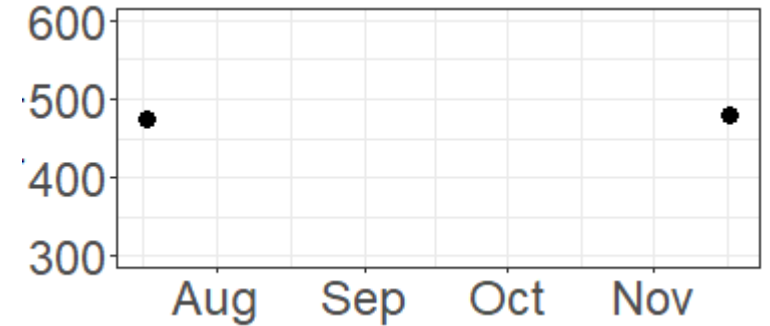
Statistical Challenges

There are several methodological challenges that need to be addressed for digital outcome measures:

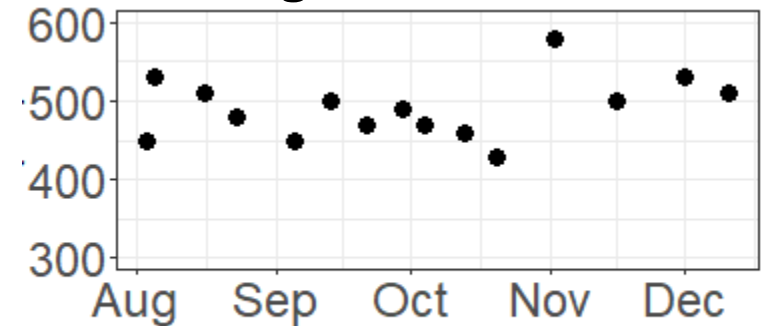
- How do you summarise the high-frequency data into a validated outcome measure?
- What is an appropriate measurement period?
- **How do you handle missing data?**

[Tackney, Carpenter and Villar (2024). *BMC Medicine*.]

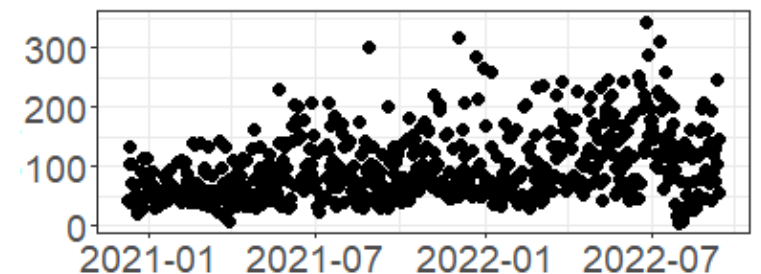
In-clinic 6MWT



Digital 6MWT



Wearable/Implantable Device



Motivating Example: PACE-UP Trial

- Baseline: wear accelerometer for 7 days
- Intervention:
 - 12-week walking intervention (postal);
 - 12-week walking intervention (nurse support);
 - Usual care
- Follow-up: wear accelerometer for 7 days



Motivating Example: PACE-UP Trial

Protocol:

- Participants were asked to wear the accelerometer from waking to sleeping each day, except when showering or swimming.
- No need to wear during sleep.



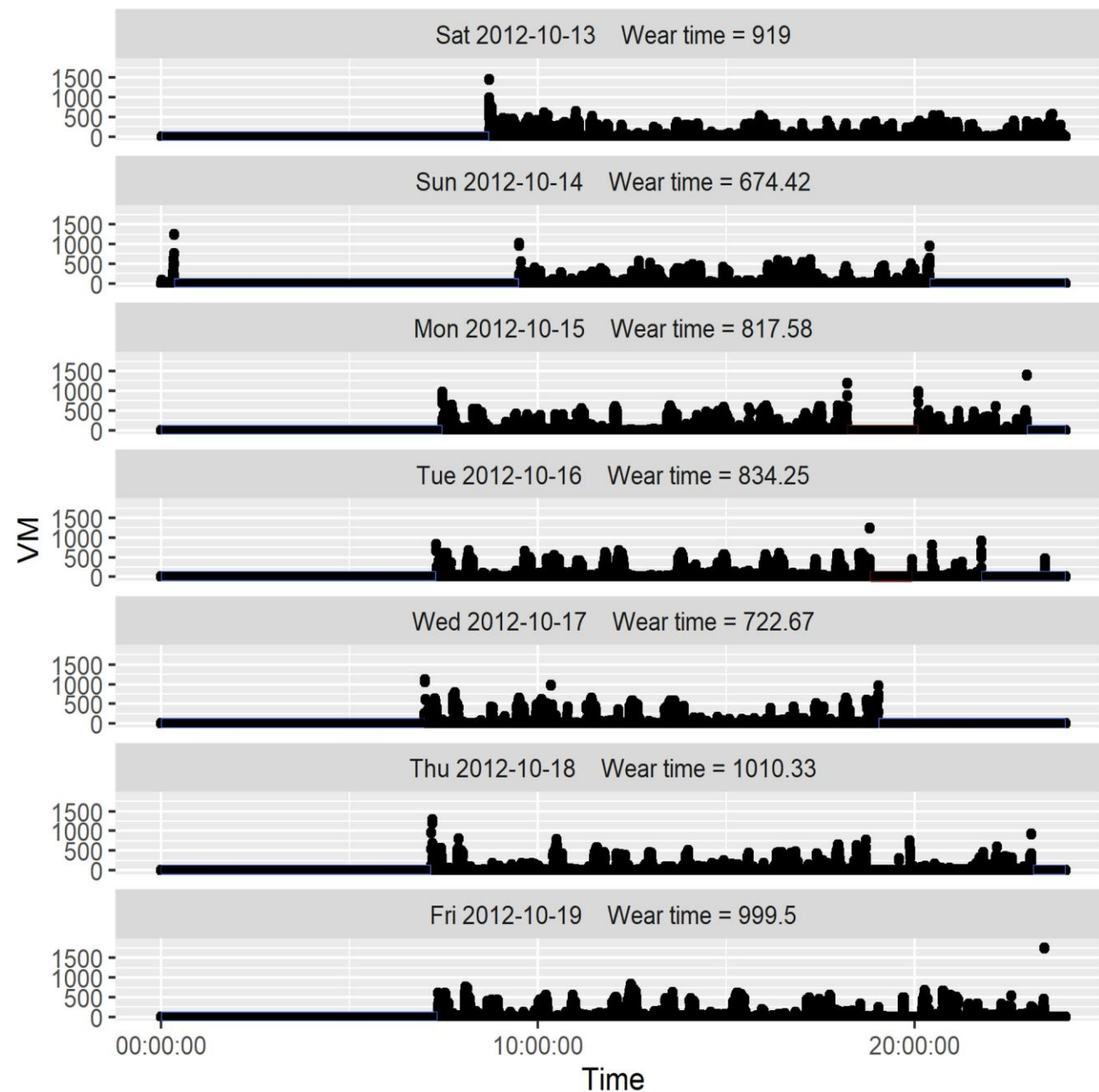
Discuss on your table (5 min)

You have three examples of 7-day epoch-level data from an accelerometer.

Do you have observations on the topic of missing data?

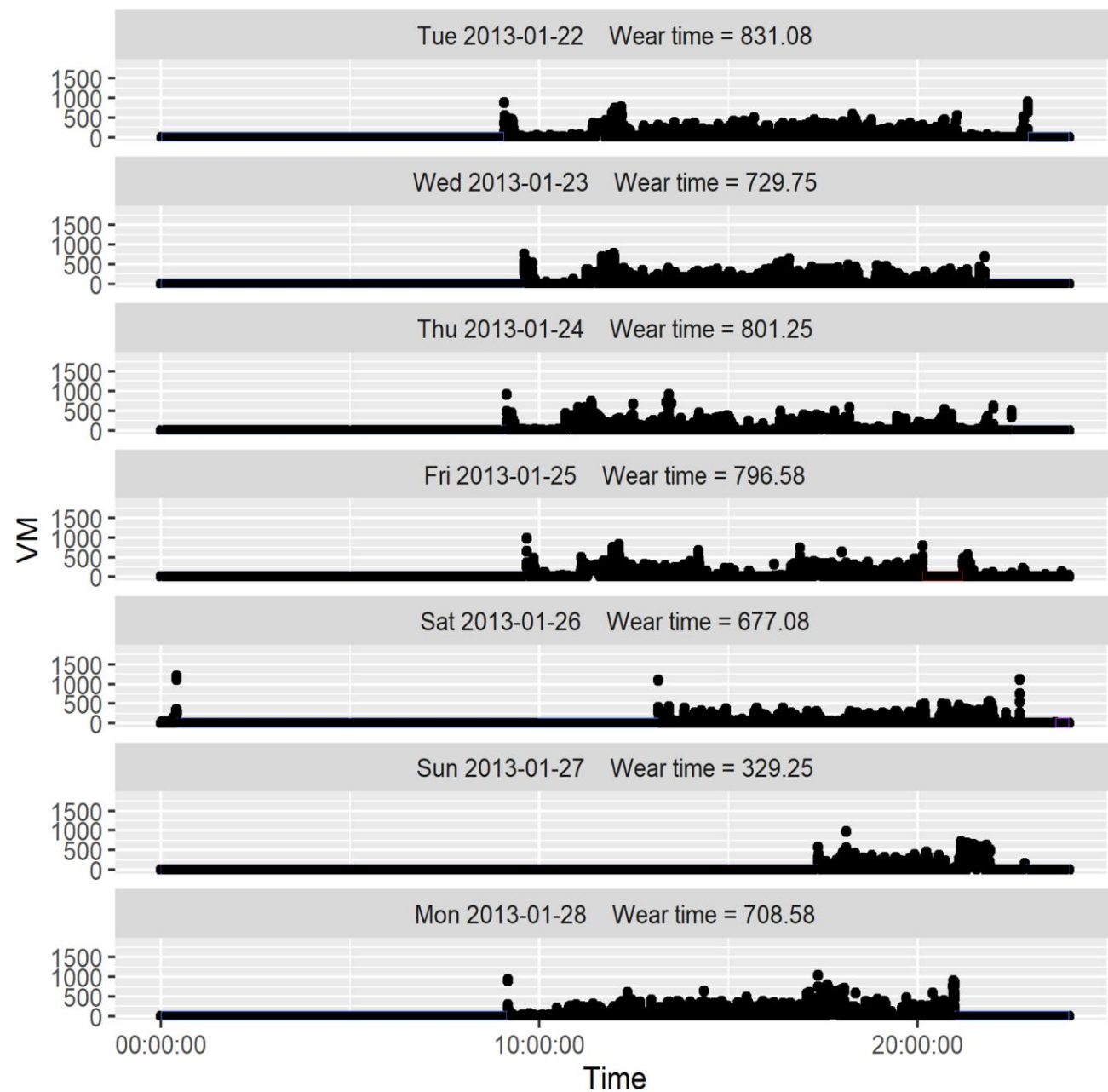
What does epoch-level data look like?

Example 1



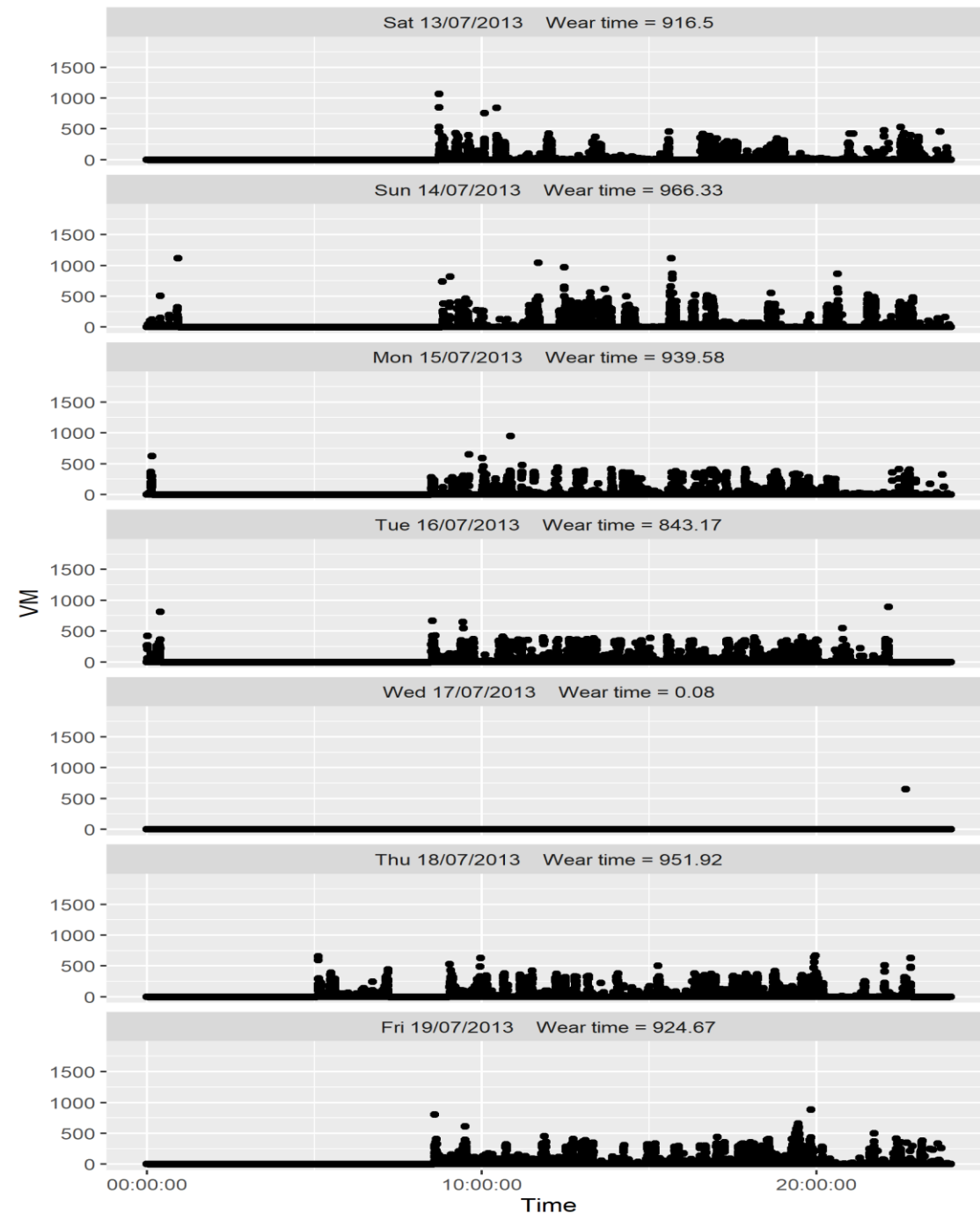
What does epoch-level data look like?

Example 2



What does epoch-level data look like?

Example 3



Missing data is common, but elusive:

- Inactivity or missing data?
- Device taken off per-protocol or missing data?
- Missing hours/days/weeks?

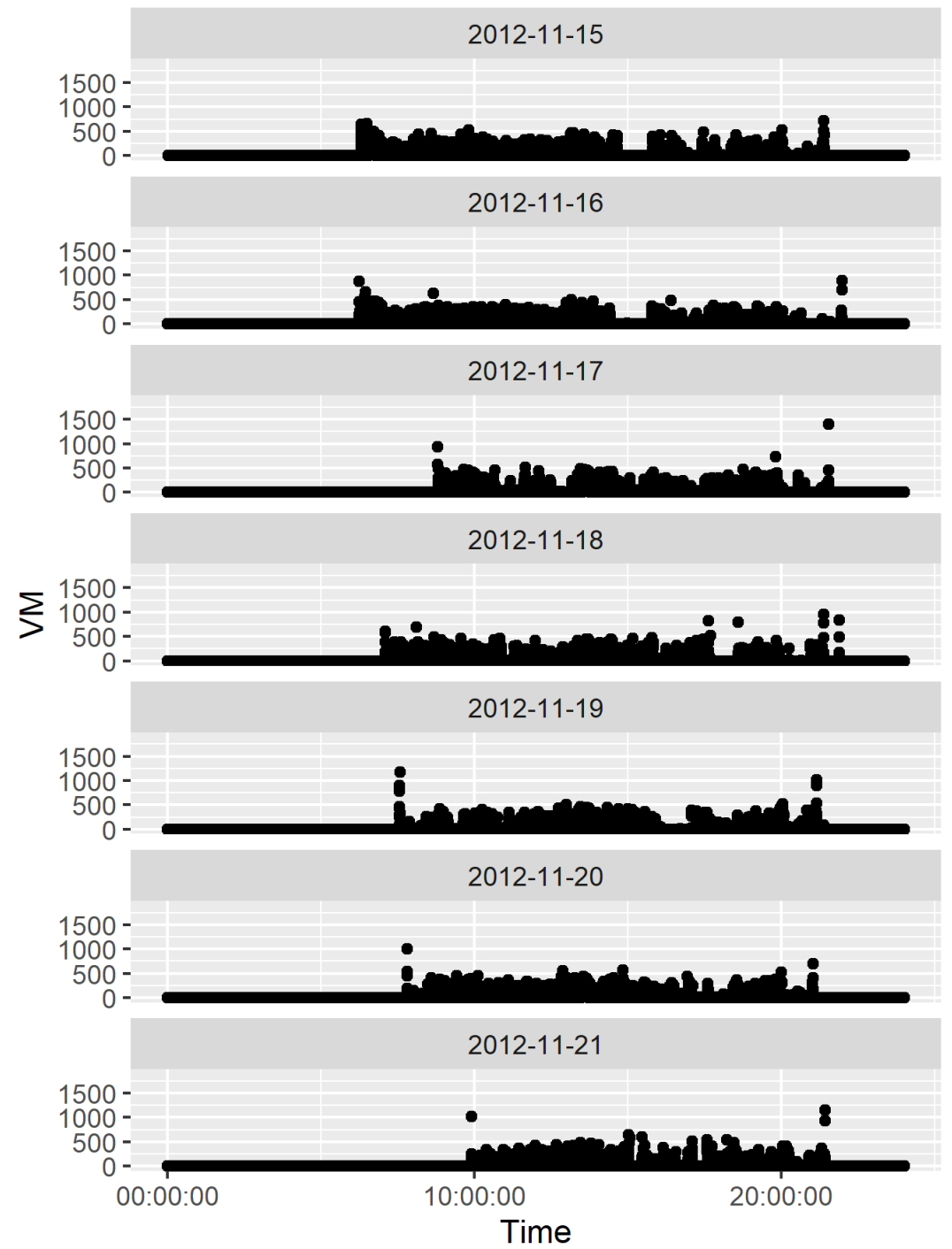
Current Practice

- Primary outcome:
average step count across the week
- Missing day if *wear time* < 540 minutes
- Simplified Analysis model:

$$\bar{y}_{i,12M} = \beta_0 + \beta_1 \bar{y}_{i,baseline} + \beta_2 I(\text{treat}_i = \text{postal}) + \beta_3 I(\text{treat}_i = \text{nurse}) + \epsilon_i$$

$$\epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Can we handle missingness better using epoch-level data?



Classifying Missingness at the Epoch-level

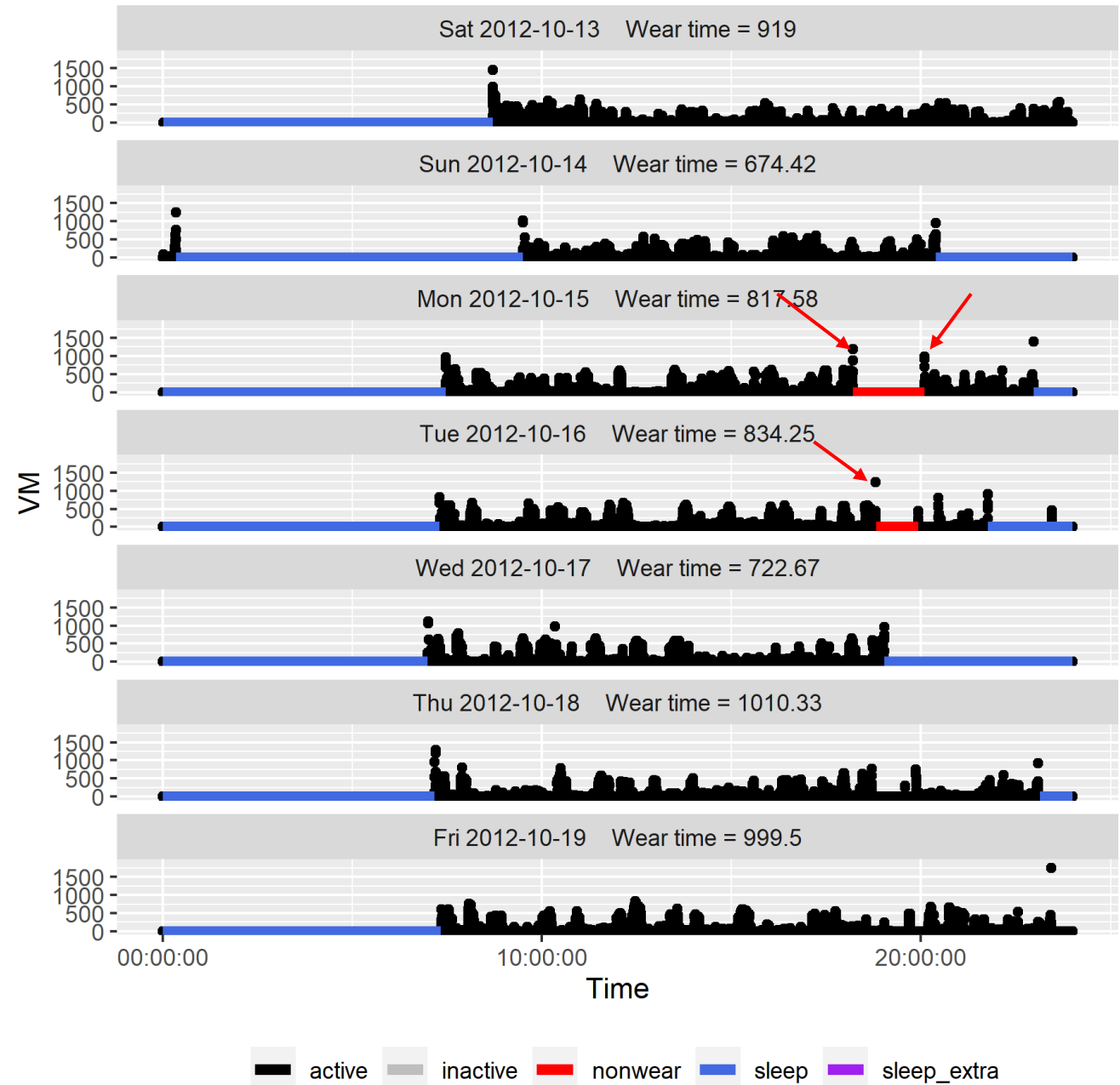


Classification (Example 1)

Active: Black dots indicate VM per epoch

Sleep: Zero-VM periods assumed to be sleep

Non-wear: High VM indicates removal of device



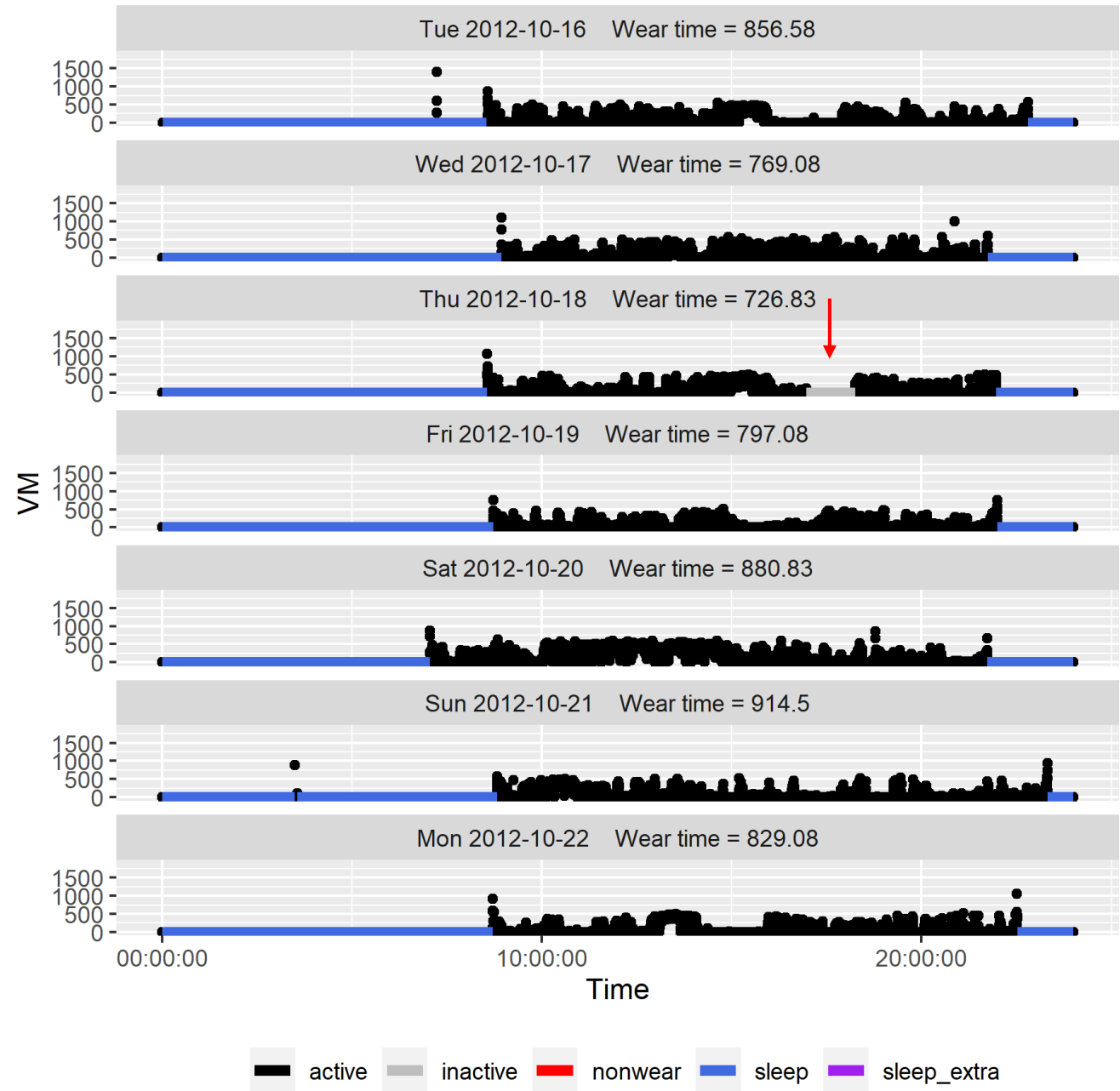
Classification

Active: Black dots indicate VM per epoch

Sleep: Zero-VM periods assumed to be sleep

Non-wear: High VM indicates removal of device

Inactive: No high VM, potentially wearing the device but inactive



Classification (Example 2)

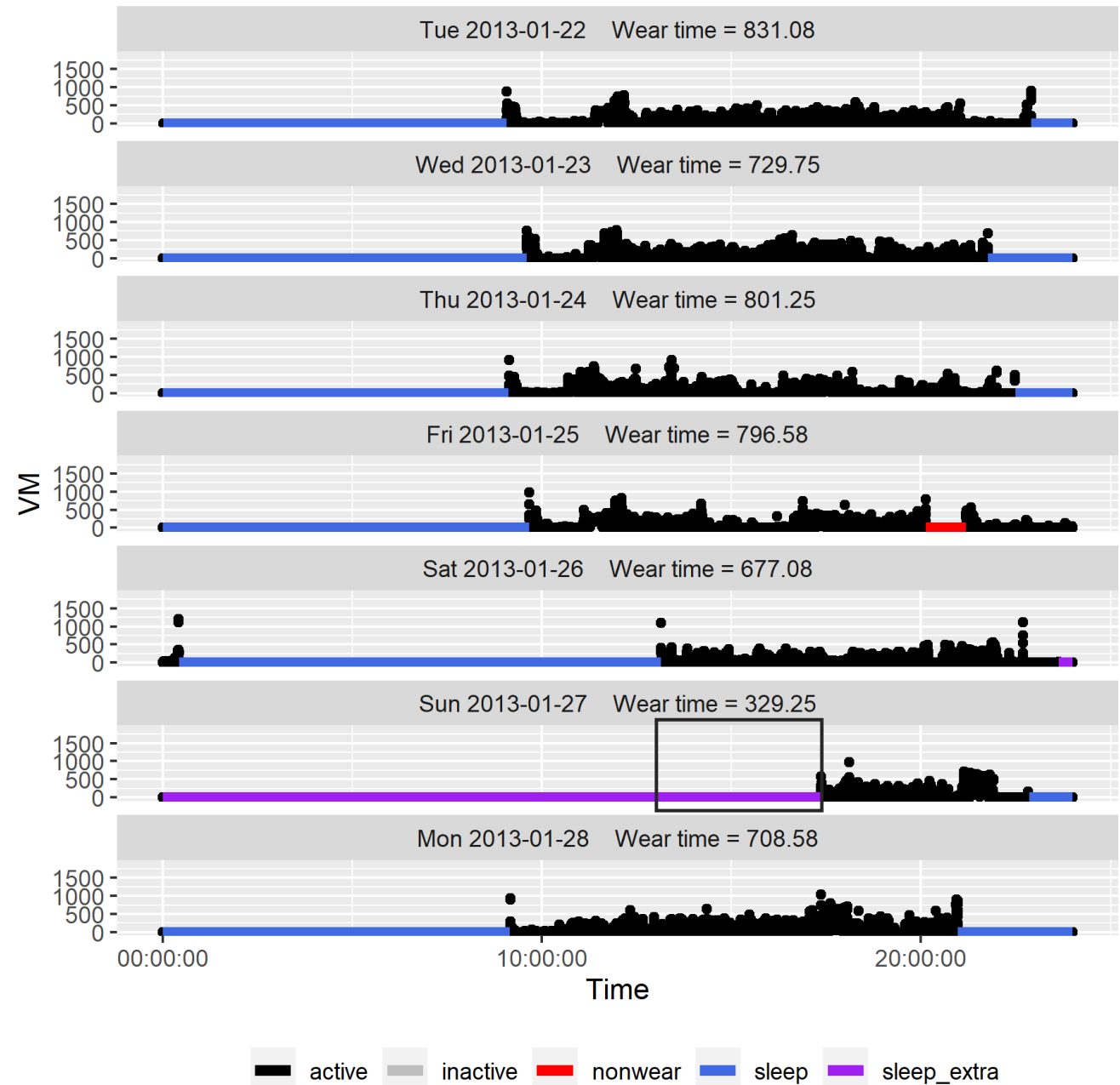
Active: Black dots indicate VM per epoch

Sleep: Zero-count periods assumed to be sleep

Non-wear: High VM indicates removal of device

Inactive: No high VM, potentially wearing the device but inactive

Sleep-extra: Participant wore the device too late/removed too early



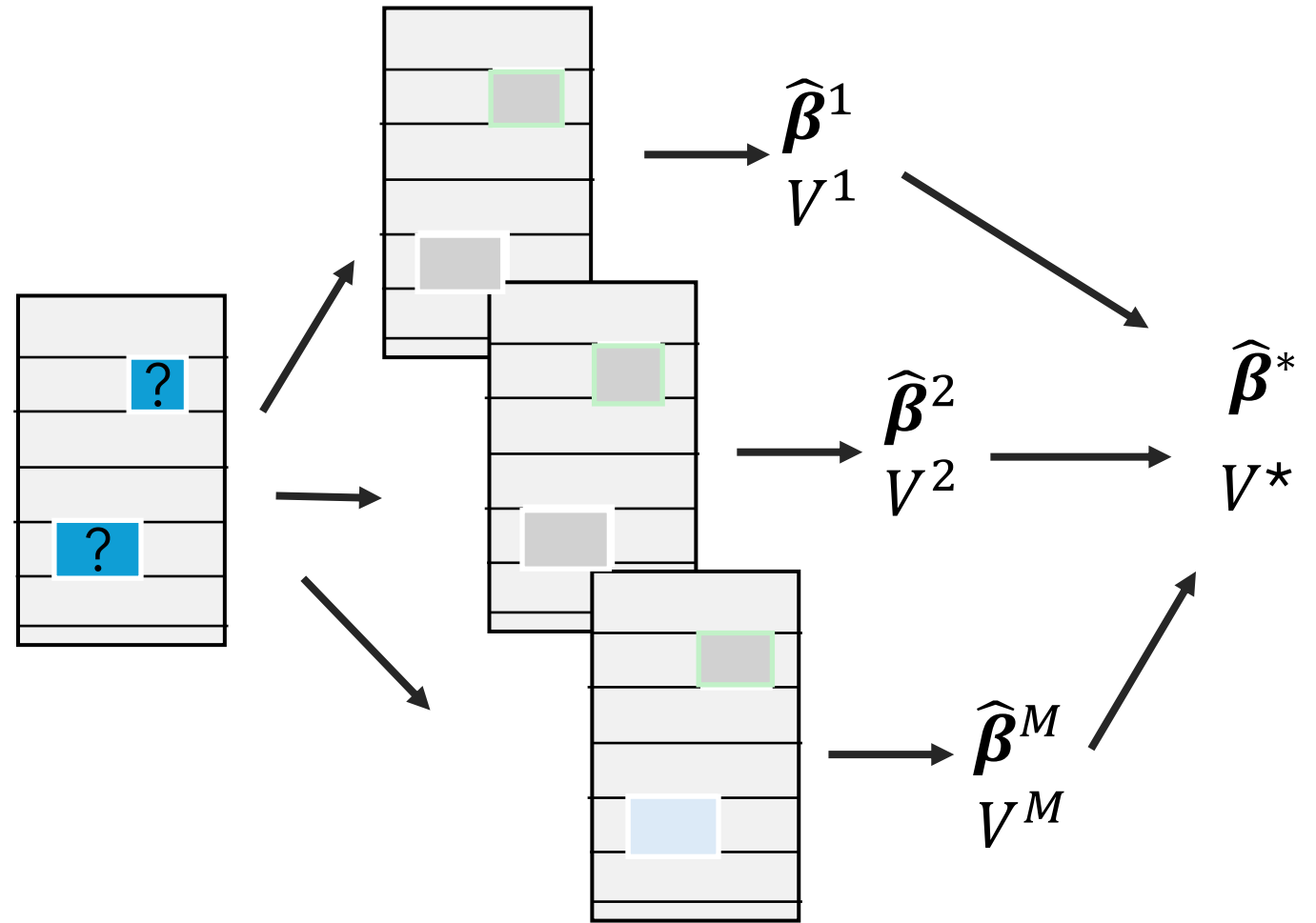
Applying Multiple Imputation at the Epoch-level



Multiple Imputation

1. Identify Missing Data
2. Replace with M plausible values under the MAR assumption
3. Obtain results of *Analysis Model* (β and variances V) for each of M complete datasets
4. Pool Results using Rubin's Rules

[Carpenter et al., 2023, *Multiple Imputation and Its Application*]



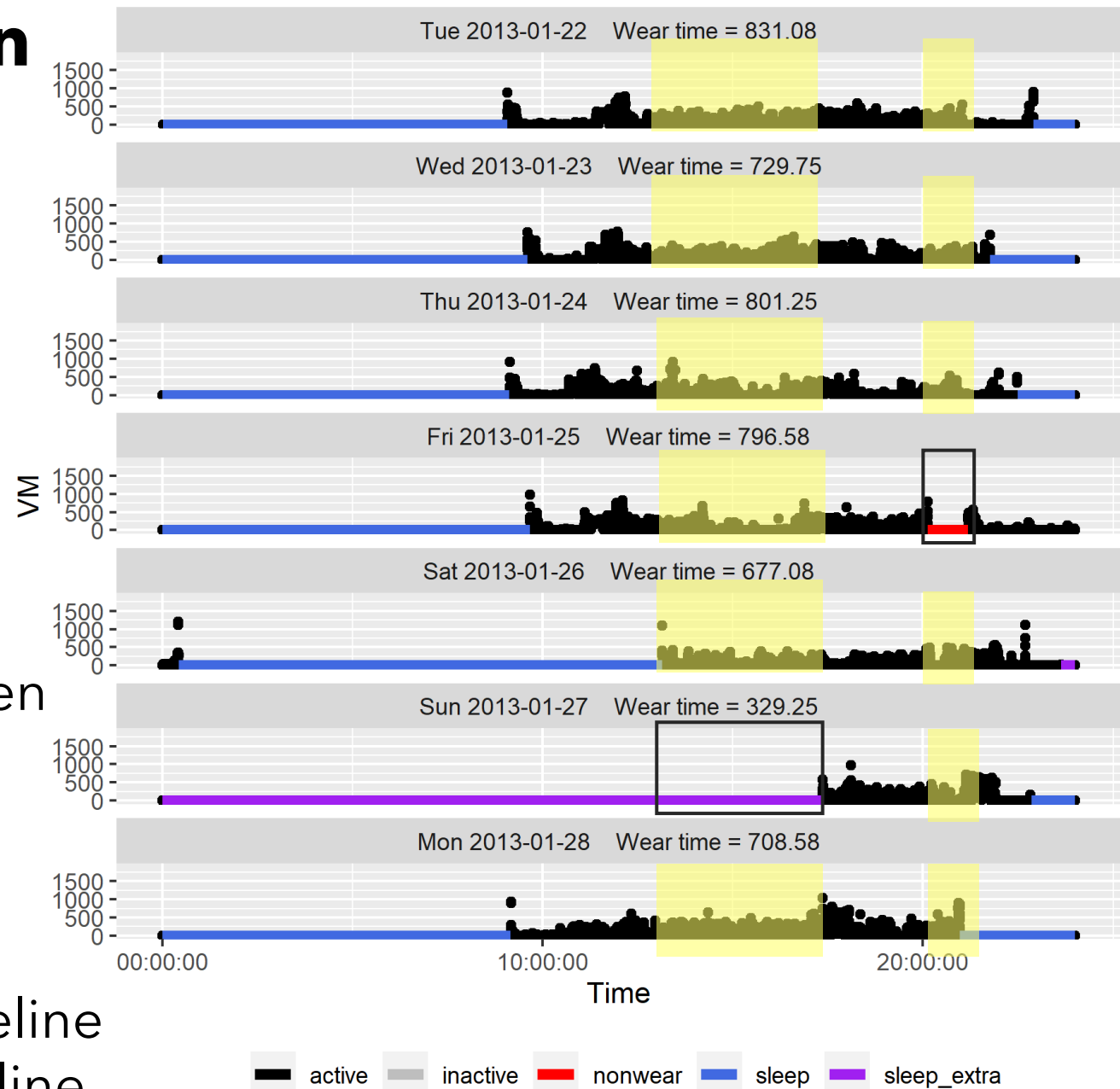
Non-parametric Imputation

Identify missing intervals

If size of self donor pool > 4 :
Impute with self donors.
Else:
Impute with non-self donors.

Match based on sex and Mahalanobis distance between demographic/activity factors, for example:

- Age
- BMI at baseline
- average step count at baseline
- average wear time at baseline



Simulation



Simulation Study

- PACE-UP trial: 1023 participants, randomized to:
 - Postal intervention;
 - Nurse-supported intervention;
 - Usual Care.
- There were 310 patients who had complete data at both baseline and 12 months (105 in postal, 101 in nurse support and 104 in control group)

Simulation Study

Focus on a realistic data generation:

- obtained complete data via bootstrap resampling from individuals with complete data (85 in each group);
- In 45% of individuals, non-wear and sleep-extra periods in 12M data were induced based on realistic patterns;
- In 2% of individuals, all data at 12M were made missing;

Simulation Study

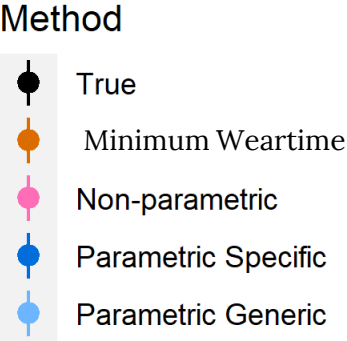
Missing data methods:

- Non-parametric epoch-level imputation approach with M=10;
- Minimum Weartime Approach
- Parametric approaches (not discussed today)

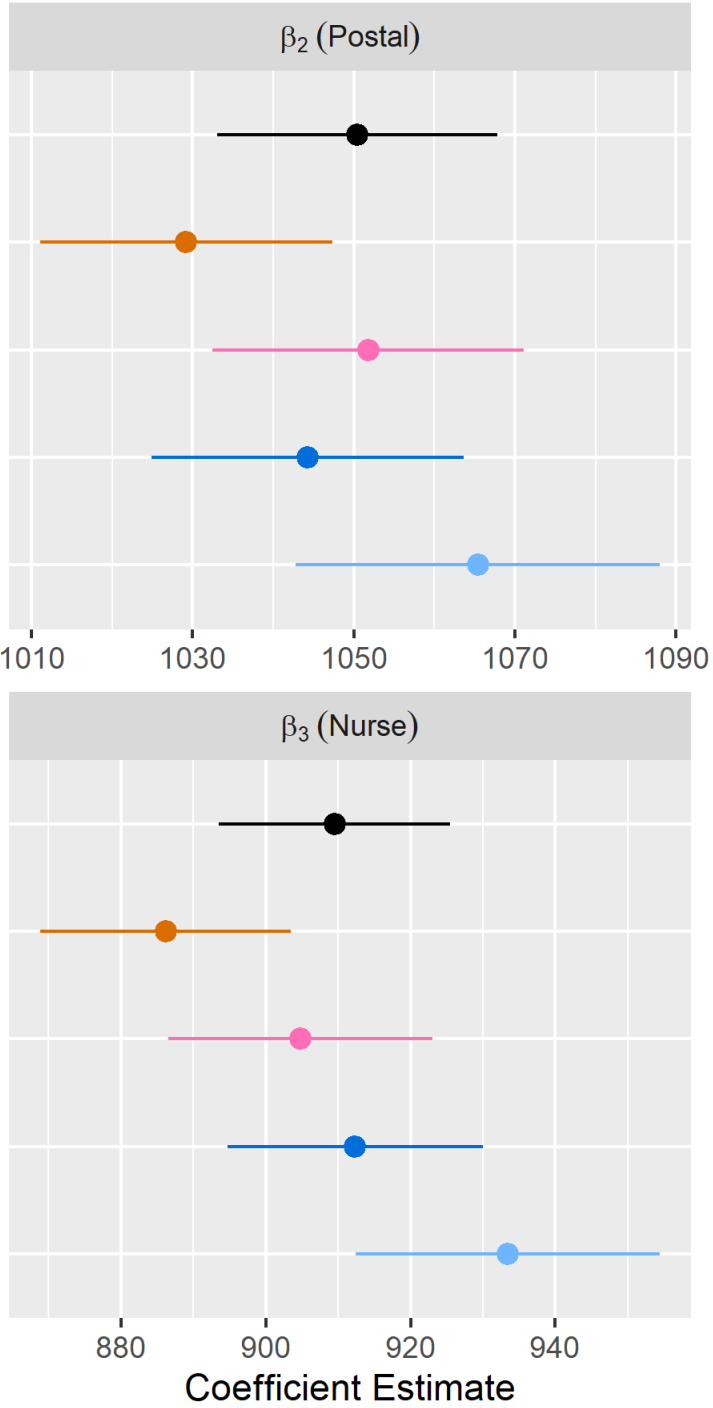
Analysis Model:

$$\bar{y}_{i,12M} = \beta_0 + \beta_1 \bar{y}_{i,baseline} + \beta_2 I(\text{treat}_i = \text{postal}) + \beta_3 I(\text{treat}_i = \text{nurse}) + \epsilon_i$$
$$\epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Results



Error bars indicate
 $1.96 \pm$ MC error



For more details...

Multiple imputation approaches for epoch-level accelerometer data in trials

Mia S Tackney¹ , Elizabeth Williamson¹ , Derek G Cook², Elizabeth Limb², Tess Harris² and James Carpenter^{1,3}

METHODOLOGY

Open Access

A framework for handling missing accelerometer outcome data in trials



Mia S. Tackney^{1*} , Derek G. Cook², Daniel Stahl⁴, Khalida Ismail⁵, Elizabeth Williamson¹ and James Carpenter^{1,3}

Considerations to address missing data when deriving clinical trial endpoints from digital health technologies

Junrui Di^{a,*}, Charmaine Demanuele^a, Anna Kettermann^b, F. Isik Karahanoglu^a, Joseph C. Cappelleri^a, Andrew Potter^b, Denise Bury^c, Jesse M. Cedarbaum^{d,e}, Bill Byrom^f



PPIE: Missing data for digital outcome measures

Discuss on your table (5 min)

Imagine bringing together a group of individuals who have participated in clinical studies with digital devices.

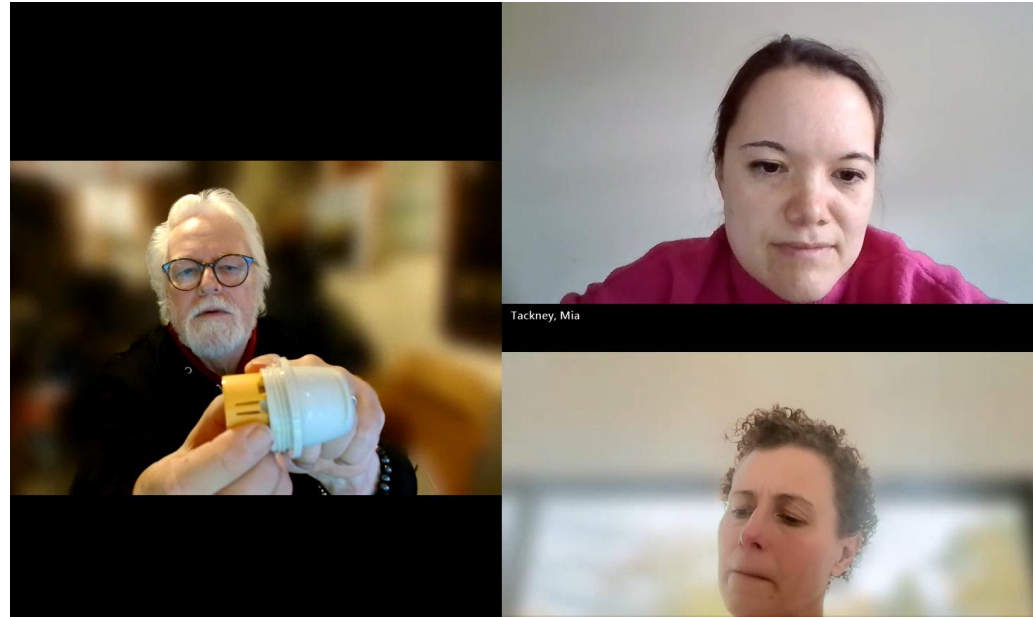
What common causes of missing data do you think would be discussed?

Major technological issues



- Features of an app that didn't work, or an app that didn't sync with a wearable device, even after seeking help from research staff;
- Could lead to frustration and drop-out.

Operational aspects of the device



- Removing devices for charging;
- Replacing parts of the device, for example the disc in a Continuous Glucose Monitor needs replacing every 15 days.

Practical aspects of daily life



- Discomfort due to hot weather;
- Swimming or hygiene activities;
- Removing device before ablution for prayer or other religious activities.

Feedback from Devices



- In some cases, lead to anxiety - for example, seeing glucose projections;
- In other cases, could lead to encouragement and mitigate missing data.

Previous experience with the Device



- Could mitigate anxiety due to seeing glucose projections;
- Could help manage discomfort, e.g. during sleep, when wearing several CGM monitors.

Informative Missingness

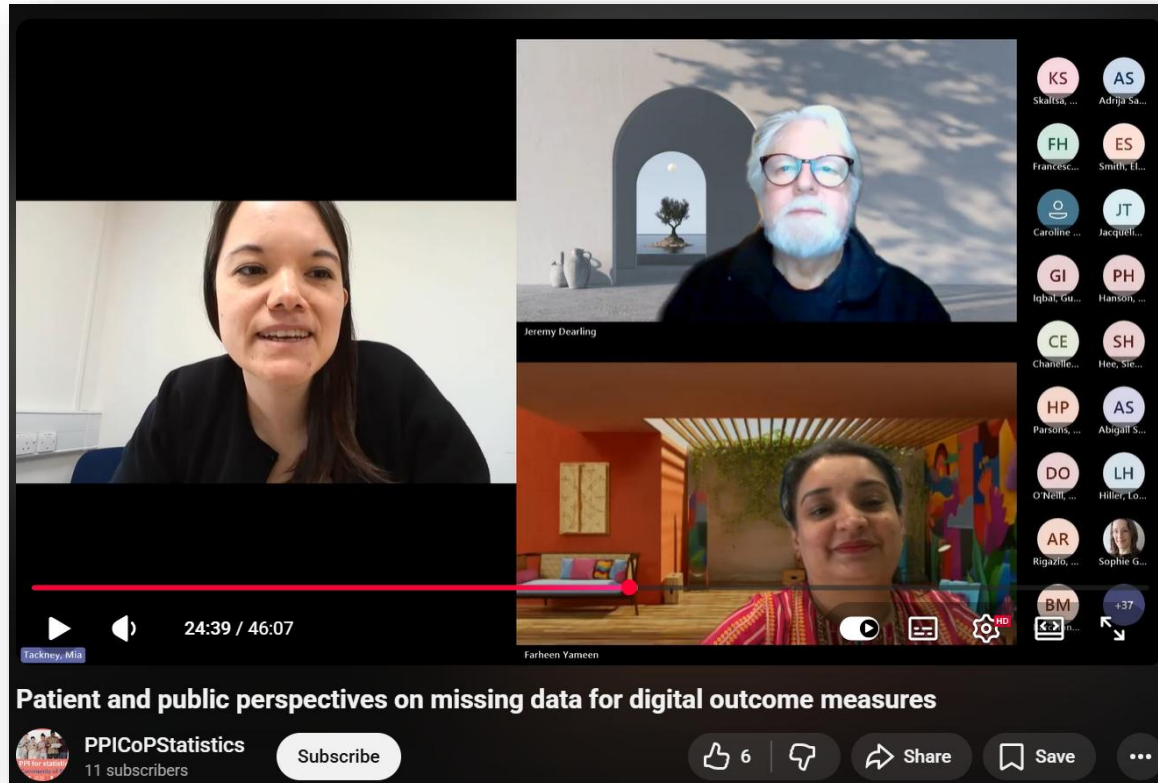
PPIE contributors thought this was important and might happen, for example:

- when symptoms worsen and engaging with studies feels a greater cognitive burden
- when participants feel they wish to hide certain aspects of their behavior

However, the PPIE contributors in the group generally wore their devices per protocol.

For more details...

PPICoPStatistics



Research Article

Can discussions with patients and the public clarify missing data mechanisms for digital outcome measures?

Mia S. Tackney, Amber Steele, Marie-Louise Zeissler, Sofía S. Villar, and 4 more

This is a preprint; it has not been peer reviewed by a journal.

<https://doi.org/10.21203/rs.3.rs-9293900/v1>
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Linking to your Hackathon task:

We've discussed how missing data might arise (**Task 1**):

- Patterns within and between individuals
- Plausible assumptions for missing data

Important differences: Your task is in

- In a different clinical context
- Focused on day-level (not epoch-level) data over a longer time scale
- Focused on identifying group-based trajectories of recovery in **Task 2** (not estimating treatment effects in a trial)